

ETHIOPIAN PAY-LATER PLATFORM

# Feature-by-Feature Functional Specification

*Detailed functional contract for every major product feature*

Ethiopia-designed • Addis Ababa-first

|               |                                                  |
|---------------|--------------------------------------------------|
| Document Code | DOC-05                                           |
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| Date          | 21 August 2026                                   |
| Prepared For  | Product • Engineering • QA • Design • Operations |
| Prepared By   | Eleanor Tefera (Fintech)                         |

CONFIDENTIAL WORKING DOCUMENT

# Document Control

| Version | Date           | Prepared By              | Status           | Summary                            |
|---------|----------------|--------------------------|------------------|------------------------------------|
| 1.0     | 21 August 2026 | Eleanor Tefera (Fintech) | Working baseline | Initial researched package release |

## Research Status Legend

|                       |                                                                    |
|-----------------------|--------------------------------------------------------------------|
| VERIFIED              | Directly supported by an official/primary source.                  |
| REPORTED              | Supported by a reputable third-party source.                       |
| INFERENCE             | Reasonable conclusion from available evidence.                     |
| ASSUMPTION            | Design assumption for planning; not presented as market fact.      |
| REQUIRES CONFIRMATION | Needs provider, bank, NBE, legal or business-partner confirmation. |

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# 1. How to Use This Document

Each feature follows the common requirement standard requested for the project. Feature IDs are stable cross-document references. Provider-dependent features are specifications for our internal behavior; they do not assert that an external provider has approved the integration.

## 2. Cross-Cutting Rules

- KYC ≠ credit approval.
- First-time credit requires CRD-004 decision.
- No merchant release when required first payment is PENDING/UNKNOWN.
- Financial writes are idempotent and ledger-backed.
- Provider-specific capabilities are configuration-gated.
- Every privileged/manual override needs reason and audit evidence.
- All features expose explicit loading/pending/error/success states.

## 3. Detailed Feature Specifications

### AUTH-001 Registration

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | AUTH-001                                                                                                                                                                                                              |
| Feature Name         | Registration                                                                                                                                                                                                          |
| Purpose              | Create a secure customer account using verified contact information.                                                                                                                                                  |
| Actors               | Customer                                                                                                                                                                                                              |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates registration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                     |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                       |

|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptance Criteria | Registration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## AUTH-002 Login

| Field                | Specification                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | AUTH-002                                                                                                                                                                                                                   |
| Feature Name         | Login                                                                                                                                                                                                                      |
| Purpose              | Authenticate returning users with role-appropriate controls.                                                                                                                                                               |
| Actors               | Customer / Merchant / Admin                                                                                                                                                                                                |
| Priority             | MVP                                                                                                                                                                                                                        |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                   |
| Main Flow            | User initiates login → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                          |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                              |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.      |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                   |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                          |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                   |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                    |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                  |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                             |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                           |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                    |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                            |
| Acceptance Criteria  | Login completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## AUTH-003 OTP

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | AUTH-003                                                                                                                                                                                 |
| Feature Name      | OTP                                                                                                                                                                                      |
| Purpose           | Verify possession of a phone number and sensitive actions.                                                                                                                               |
| Actors            | Customer / Merchant                                                                                                                                                                      |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates otp → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                 |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review;                                                                                                                     |

|                      |                                                                                                                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | customer cancels before irreversible financial action; provider fallback only when approved.                                                                                                                             |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                            |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.    |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                 |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                        |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                 |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                  |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                           |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                         |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                  |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                          |
| Acceptance Criteria  | OTP completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CUS-001 Customer Profile

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CUS-001                                                                                                                                                                                                               |
| Feature Name         | Customer Profile                                                                                                                                                                                                      |
| Purpose              | Maintain identity, contact, preferences and account status.                                                                                                                                                           |
| Actors               | Customer                                                                                                                                                                                                              |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates customer profile → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |

|                     |                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria | Customer Profile completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## KYC-001 Identity Verification

| Field                | Specification                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | KYC-001                                                                                                                                                                                                                                    |
| Feature Name         | Identity Verification                                                                                                                                                                                                                      |
| Purpose              | Verify identity before Pay-Later eligibility.                                                                                                                                                                                              |
| Actors               | Customer / Compliance                                                                                                                                                                                                                      |
| Priority             | MVP                                                                                                                                                                                                                                        |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                   |
| Main Flow            | User initiates identity verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                          |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                              |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                      |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                   |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                          |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                   |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                    |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                  |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                             |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                           |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                    |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                            |
| Acceptance Criteria  | Identity Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## KYC-002 Fayda Verification

| Field         | Specification                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID    | KYC-002                                                                                                                                                   |
| Feature Name  | Fayda Verification                                                                                                                                        |
| Purpose       | Use Fayda/accepted digital identity only if legally and technically available.                                                                            |
| Actors        | Customer / Compliance                                                                                                                                     |
| Priority      | Should                                                                                                                                                    |
| Preconditions | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled |



|                      |                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | when configured and confirmed.                                                                                                                                                                                                          |
| Main Flow            | User initiates fayda verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                       |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                           |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                   |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                       |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                 |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                               |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                          |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                        |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                 |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                         |
| Acceptance Criteria  | Fayda Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-001 Merchant Registration

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | MER-001                                                                                                                                                                                                               |
| Feature Name         | Merchant Registration                                                                                                                                                                                                 |
| Purpose              | Create merchant account and capture business information.                                                                                                                                                             |
| Actors               | Merchant Owner                                                                                                                                                                                                        |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates merchant registration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                            |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |

|                     |                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                  |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                             |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                           |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                    |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                            |
| Acceptance Criteria | Merchant Registration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-002 Business Verification

| Field                | Specification                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | MER-002                                                                                                                                                                                                                                    |
| Feature Name         | Business Verification                                                                                                                                                                                                                      |
| Purpose              | Verify merchant legitimacy before BNPL transactions.                                                                                                                                                                                       |
| Actors               | Merchant / Compliance                                                                                                                                                                                                                      |
| Priority             | MVP                                                                                                                                                                                                                                        |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                   |
| Main Flow            | User initiates business verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                          |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                              |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                      |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                   |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                          |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                   |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                    |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                  |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                             |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                           |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                    |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                            |
| Acceptance Criteria  | Business Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-003 Branch Management

| Field        | Specification     |
|--------------|-------------------|
| Feature ID   | MER-003           |
| Feature Name | Branch Management |

|                      |                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose              | Manage merchant branches and operating status.                                                                                                                                                                                         |
| Actors               | Merchant Owner / Manager                                                                                                                                                                                                               |
| Priority             | MVP                                                                                                                                                                                                                                    |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                               |
| Main Flow            | User initiates branch management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                      |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                          |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                  |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                               |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                      |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                               |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                              |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                         |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                       |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                        |
| Acceptance Criteria  | Branch Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-004 Employee Management

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | MER-004                                                                                                                                                                                                               |
| Feature Name         | Employee Management                                                                                                                                                                                                   |
| Purpose              | Invite, role and deactivate merchant employees.                                                                                                                                                                       |
| Actors               | Merchant Owner / Manager                                                                                                                                                                                              |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates employee management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                              |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state                                                                                                                                            |

|                     |                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | transition; duplicate/idempotency checks; risk rules where applicable.                                                                                                                                                                   |
| User Interface      | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                 |
| Notifications       | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                  |
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                           |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                         |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                  |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                          |
| Acceptance Criteria | Employee Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-005 — Product Management

| Field                | Specification                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | MER-005                                                                                                                                                                                                                                 |
| Feature Name         | Product Management                                                                                                                                                                                                                      |
| Purpose              | Maintain optional catalog for merchants that need it.                                                                                                                                                                                   |
| Actors               | Merchant                                                                                                                                                                                                                                |
| Priority             | Should                                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                |
| Main Flow            | User initiates product management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                       |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                           |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                   |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                       |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                 |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                               |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                          |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                        |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                 |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                         |
| Acceptance Criteria  | Product Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## MER-006 — Quick Sale

| Field                | Specification                                                                                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | MER-006                                                                                                                                                                                                                         |
| Feature Name         | Quick Sale                                                                                                                                                                                                                      |
| Purpose              | Create a sale from description + amount without full catalog/POS.                                                                                                                                                               |
| Actors               | Merchant                                                                                                                                                                                                                        |
| Priority             | MVP                                                                                                                                                                                                                             |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                        |
| Main Flow            | User initiates quick sale → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                               |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                   |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.           |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                        |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                               |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                        |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                         |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                       |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                  |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                         |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                 |
| Acceptance Criteria  | Quick Sale completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## QR-001 — Merchant QR

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | QR-001                                                                                                                                                                                   |
| Feature Name      | Merchant QR                                                                                                                                                                              |
| Purpose           | Start a merchant-linked checkout by QR.                                                                                                                                                  |
| Actors            | Merchant / Customer                                                                                                                                                                      |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates merchant qr → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.         |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                        |
| Failure Flows     | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                            |
| Business Rules    | Rules are configuration/version controlled; no financial state changes without                                                                                                           |

|                      |                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                                                                                           |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                         |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                         |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                          |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                        |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                   |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                 |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                          |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                  |
| Acceptance Criteria  | Merchant QR completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## QR-002 — Transaction QR

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | QR-002                                                                                                                                                                                                                |
| Feature Name         | Transaction QR                                                                                                                                                                                                        |
| Purpose              | Open a specific sale safely using short-lived QR.                                                                                                                                                                     |
| Actors               | Merchant / Customer                                                                                                                                                                                                   |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates transaction qr → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                   |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                       |

|                     |                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptance Criteria | Transaction QR completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## DISC-001 — Merchant Discovery

| Field                | Specification                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | DISC-001                                                                                                                                                                                                                                |
| Feature Name         | Merchant Discovery                                                                                                                                                                                                                      |
| Purpose              | Find participating merchants and categories.                                                                                                                                                                                            |
| Actors               | Customer                                                                                                                                                                                                                                |
| Priority             | Should                                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                |
| Main Flow            | User initiates merchant discovery → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                       |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                           |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                   |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                       |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                 |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                               |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                          |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                        |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                 |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                         |
| Acceptance Criteria  | Merchant Discovery completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CRD-001 — Customer Eligibility

| Field             | Specification                                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | CRD-001                                                                                                                                                                                   |
| Feature Name      | Customer Eligibility                                                                                                                                                                      |
| Purpose           | Determine whether Pay-Later is available responsibly.                                                                                                                                     |
| Actors            | Customer / Risk                                                                                                                                                                           |
| Priority          | MVP                                                                                                                                                                                       |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.  |
| Main Flow         | User initiates customer eligibility → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications. |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only                                        |

|                      |                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | when approved.                                                                                                                                                                                                                            |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria  | Customer Eligibility completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CRD-002 — Credit Limit

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CRD-002                                                                                                                                                                                                               |
| Feature Name         | Credit Limit                                                                                                                                                                                                          |
| Purpose              | Manage approved, available, reserved and restricted exposure.                                                                                                                                                         |
| Actors               | Customer / Risk                                                                                                                                                                                                       |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates credit limit → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                     |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID,                                                                                                                                           |



|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | policy/config version and timestamp.                                                                                                                                                                                              |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                           |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                   |
| Acceptance Criteria | Credit Limit completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CRD-003 — Transaction Decision

| Field                | Specification                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CRD-003                                                                                                                                                                                                                                   |
| Feature Name         | Transaction Decision                                                                                                                                                                                                                      |
| Purpose              | Reassess each purchase even when a customer has a limit.                                                                                                                                                                                  |
| Actors               | Customer / Risk                                                                                                                                                                                                                           |
| Priority             | MVP                                                                                                                                                                                                                                       |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                  |
| Main Flow            | User initiates transaction decision → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria  | Transaction Decision completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CRD-004 — First-Time Eligibility

| Field         | Specification                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID    | CRD-004                                                                                                                                                   |
| Feature Name  | First-Time Eligibility                                                                                                                                    |
| Purpose       | Approve, restrict, request information, review or decline first use without relying on internal history.                                                  |
| Actors        | Customer / Risk / Credit Partner                                                                                                                          |
| Priority      | MVP                                                                                                                                                       |
| Preconditions | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled |

|                            |                                                                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | when configured and confirmed.                                                                                                                                                                                                              |
| Main Flow                  | User initiates first-time eligibility → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows          | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                           |
| Failure Flows              | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                               |
| Business Rules             | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                       |
| Required Information       | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                    |
| Validation                 | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                           |
| User Interface             | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                    |
| Notifications              | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                     |
| Security                   | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                   |
| Financial Impact           | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                              |
| Audit Requirements         | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                            |
| Data Requirements          | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                     |
| Dependencies               | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                             |
| Acceptance Criteria        | First-Time Eligibility completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |
|                            |                                                                                                                                                                                                                                             |
| APPROVED_STARTER_LIMIT     | Reusable conservative starter exposure.                                                                                                                                                                                                     |
| APPROVED_TRANSACTION_ONLY  | Only current transaction approved.                                                                                                                                                                                                          |
| APPROVED_WITH_DOWN_PAYMENT | Higher customer contribution reduces financed exposure.                                                                                                                                                                                     |
| MORE_INFORMATION_REQUIRED  | No decision until defined evidence supplied.                                                                                                                                                                                                |
| MANUAL_REVIEW              | Human/partner review required.                                                                                                                                                                                                              |
| DECLINED                   | No financing.                                                                                                                                                                                                                               |
| TEMPORARILY_BLOCKED        | Security/fraud condition prevents decision.                                                                                                                                                                                                 |

## CRD-005 — Limit Graduation

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | CRD-005                                                                                                                                                                                  |
| Feature Name      | Limit Graduation                                                                                                                                                                         |
| Purpose           | Review future limits using repayment and affordability evidence.                                                                                                                         |
| Actors            | Risk                                                                                                                                                                                     |
| Priority          | Should                                                                                                                                                                                   |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates limit graduation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.    |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                        |
| Failure Flows     | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                            |
| Business Rules    | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit                            |

|                      |                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | and payment decisions are not inferred from UI success.                                                                                                                                                                               |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria  | Limit Graduation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CRD-006 — Offer Engine

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CRD-006                                                                                                                                                                                                               |
| Feature Name         | Offer Engine                                                                                                                                                                                                          |
| Purpose              | Return responsible alternatives such as higher down payment or lower financed amount.                                                                                                                                 |
| Actors               | Customer / Risk                                                                                                                                                                                                       |
| Priority             | Should                                                                                                                                                                                                                |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates offer engine → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                     |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                       |

|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptance Criteria | Offer Engine completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## BNPL-001 — Pay-Later Purchase

| Field                | Specification                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | BNPL-001                                                                                                                                                                                                                                |
| Feature Name         | Pay-Later Purchase                                                                                                                                                                                                                      |
| Purpose              | Create an approved installment purchase.                                                                                                                                                                                                |
| Actors               | Customer / Merchant                                                                                                                                                                                                                     |
| Priority             | MVP                                                                                                                                                                                                                                     |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                |
| Main Flow            | User initiates pay-later purchase → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                       |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                           |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                   |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                       |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                 |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                               |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                          |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                        |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                 |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                         |
| Acceptance Criteria  | Pay-Later Purchase completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## BNPL-002 — Installment Plans

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | BNPL-002                                                                                                                                                                                 |
| Feature Name      | Installment Plans                                                                                                                                                                        |
| Purpose           | Present configurable schedules and total-payable disclosure.                                                                                                                             |
| Actors            | Customer                                                                                                                                                                                 |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates installment plans → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.   |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only                                       |

|                      |                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | when approved.                                                                                                                                                                                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                          |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                  |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                               |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                      |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                               |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                              |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                         |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                       |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                        |
| Acceptance Criteria  | Installment Plans completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## BNPL-003 — First Payment

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | BNPL-003                                                                                                                                                                                                              |
| Feature Name         | First Payment                                                                                                                                                                                                         |
| Purpose              | Collect required initial payment before activating financing/release.                                                                                                                                                 |
| Actors               | Customer                                                                                                                                                                                                              |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates first payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                    |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID,                                                                                                                                           |

|                     |                                                                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | policy/config version and timestamp.                                                                                                                                                                                               |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                            |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                    |
| Acceptance Criteria | First Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

Specific acceptance: for a 2,000 ETB illustrative Pay-in-4 plan, the 500 ETB first payment must be SUCCESS before the remaining 1,500 ETB financed exposure is activated and merchant release is confirmed. PENDING/UNKNOWN remains on hold.

## BNPL-004 — Flexible Down Payment

| Field                | Specification                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | BNPL-004                                                                                                                                                                                                                                   |
| Feature Name         | Flexible Down Payment                                                                                                                                                                                                                      |
| Purpose              | Allow a larger upfront payment without increasing approved credit.                                                                                                                                                                         |
| Actors               | Customer                                                                                                                                                                                                                                   |
| Priority             | Should                                                                                                                                                                                                                                     |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                   |
| Main Flow            | User initiates flexible down payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                          |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                              |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                      |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                   |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                          |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                   |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                    |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                  |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                             |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                           |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                    |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                            |
| Acceptance Criteria  | Flexible Down Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-001 — Payment Method Selection

| Field        | Specification                                    |
|--------------|--------------------------------------------------|
| Feature ID   | PAY-001                                          |
| Feature Name | Payment Method Selection                         |
| Purpose      | Choose an available, supported repayment method. |
| Actors       | Customer                                         |

|                      |                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority             | MVP                                                                                                                                                                                                                                           |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                      |
| Main Flow            | User initiates payment method selection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                             |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                                 |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                         |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                      |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                             |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                      |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                       |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                     |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                                |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                              |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                       |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                               |
| Acceptance Criteria  | Payment Method Selection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-002 — Telebirr Payment

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-002                                                                                                                                                                                                               |
| Feature Name         | Telebirr Payment                                                                                                                                                                                                      |
| Purpose              | Initiate/confirm Telebirr payment using verified public/commercial capability.                                                                                                                                        |
| Actors               | Customer                                                                                                                                                                                                              |
| Priority             | MVP*                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates telebirr payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |

|                     |                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notifications       | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria | Telebirr Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

### PAY-003 — CBE Birr Payment

| Field                | Specification                                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-003                                                                                                                                                                                                                               |
| Feature Name         | CBE Birr Payment                                                                                                                                                                                                                      |
| Purpose              | Support CBE Birr only through confirmed provider/gateway arrangement.                                                                                                                                                                 |
| Actors               | Customer                                                                                                                                                                                                                              |
| Priority             | MVP*                                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                              |
| Main Flow            | User initiates cbe birr payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                 |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria  | CBE Birr Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |



Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

## PAY-004 — M-PESA Ethiopia Payment

| Field                | Specification                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-004                                                                                                                                                                                                                                      |
| Feature Name         | M-PESA Ethiopia Payment                                                                                                                                                                                                                      |
| Purpose              | Support M-PESA Ethiopia through confirmed provider/gateway arrangement.                                                                                                                                                                      |
| Actors               | Customer                                                                                                                                                                                                                                     |
| Priority             | MVP*                                                                                                                                                                                                                                         |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                     |
| Main Flow            | User initiates m-pesa ethiopia payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                            |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                                |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                        |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                     |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                            |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                     |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                      |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                    |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                               |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                             |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                      |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                              |
| Acceptance Criteria  | M-PESA Ethiopia Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

## PAY-005 — Bank Payment

| Field         | Specification                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID    | PAY-005                                                                                                                                                                                  |
| Feature Name  | Bank Payment                                                                                                                                                                             |
| Purpose       | Accept supported bank payment rails with reference/reconciliation controls.                                                                                                              |
| Actors        | Customer / Finance                                                                                                                                                                       |
| Priority      | Should                                                                                                                                                                                   |
| Preconditions | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main Flow            | User initiates bank payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                 |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                     |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.             |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                          |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                 |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                          |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                           |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                         |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                    |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                  |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                           |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                   |
| Acceptance Criteria  | Bank Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-006 — Payment Gateway

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-006                                                                                                                                                                                                               |
| Feature Name         | Payment Gateway                                                                                                                                                                                                       |
| Purpose              | Use licensed gateway to reach multiple Ethiopian payment methods.                                                                                                                                                     |
| Actors               | Customer / Platform                                                                                                                                                                                                   |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates payment gateway → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                  |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |

|                     |                                                                                                                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                       |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                     |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                              |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                      |
| Acceptance Criteria | Payment Gateway completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-007 — Payment Confirmation

| Field                | Specification                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-007                                                                                                                                                                                                                                   |
| Feature Name         | Payment Confirmation                                                                                                                                                                                                                      |
| Purpose              | Authoritatively confirm provider outcome and post financial effects once.                                                                                                                                                                 |
| Actors               | Platform                                                                                                                                                                                                                                  |
| Priority             | MVP                                                                                                                                                                                                                                       |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                  |
| Main Flow            | User initiates payment confirmation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria  | Payment Confirmation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-008 — Payment Retry

| Field        | Specification                                           |
|--------------|---------------------------------------------------------|
| Feature ID   | PAY-008                                                 |
| Feature Name | Payment Retry                                           |
| Purpose      | Retry safely using idempotency without double charging. |
| Actors       | Customer / Platform                                     |

|                      |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority             | MVP                                                                                                                                                                                                                                |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                           |
| Main Flow            | User initiates payment retry → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                  |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                      |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.              |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                           |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                  |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                           |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                            |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                          |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                     |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                   |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                            |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                    |
| Acceptance Criteria  | Payment Retry completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-009 — Automatic Repayment

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-009                                                                                                                                                                                                               |
| Feature Name         | Automatic Repayment                                                                                                                                                                                                   |
| Purpose              | Auto-collect only with explicit mandate/provider support.                                                                                                                                                             |
| Actors               | Customer / Platform                                                                                                                                                                                                   |
| Priority             | Later                                                                                                                                                                                                                 |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates automatic repayment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                              |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |

|                     |                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notifications       | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                  |
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                           |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                         |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                  |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                          |
| Acceptance Criteria | Automatic Repayment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-010 — Payment Reminders

| Field                | Specification                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-010                                                                                                                                                                                                                                |
| Feature Name         | Payment Reminders                                                                                                                                                                                                                      |
| Purpose              | Notify before/during overdue according to policy.                                                                                                                                                                                      |
| Actors               | Customer                                                                                                                                                                                                                               |
| Priority             | MVP                                                                                                                                                                                                                                    |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                               |
| Main Flow            | User initiates payment reminders → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                      |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                          |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                  |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                               |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                      |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                               |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                              |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                         |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                       |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                        |
| Acceptance Criteria  | Payment Reminders completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## PAY-011 — Failed / Pending Payment

| Field                | Specification                                                                                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | PAY-011                                                                                                                                                                                                                                       |
| Feature Name         | Failed / Pending Payment                                                                                                                                                                                                                      |
| Purpose              | Handle FAILED separately from PENDING/UNKNOWN.                                                                                                                                                                                                |
| Actors               | Platform / Support                                                                                                                                                                                                                            |
| Priority             | MVP                                                                                                                                                                                                                                           |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                      |
| Main Flow            | User initiates failed / pending payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                             |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                                 |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                         |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                      |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                             |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                      |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                       |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                     |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                                |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                              |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                       |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                               |
| Acceptance Criteria  | Failed / Pending Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

State rule: a timeout or lost callback creates PENDING/UNKNOWN and a reconciliation job; it is not converted to FAILED merely because the app did not receive an immediate success response.

## COL-001 — Overdue Payments

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | COL-001                                                                                                                                                                                  |
| Feature Name      | Overdue Payments                                                                                                                                                                         |
| Purpose           | Apply reminders, restrictions, review and responsible collections.                                                                                                                       |
| Actors            | Customer / Risk / Support                                                                                                                                                                |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates overdue payments → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.    |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review;                                                                                                                     |

|                      |                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | customer cancels before irreversible financial action; provider fallback only when approved.                                                                                                                                          |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                 |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria  | Overdue Payments completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## COL-002 — Restructuring / Hardship

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | COL-002                                                                                                                                                                                                               |
| Feature Name         | Restructuring / Hardship                                                                                                                                                                                              |
| Purpose              | Support approved repayment arrangements/restructuring.                                                                                                                                                                |
| Actors               | Customer / Support / Credit Partner                                                                                                                                                                                   |
| Priority             | Later                                                                                                                                                                                                                 |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates restructuring / hardship → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                         |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |

|                     |                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                              |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                       |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                               |
| Acceptance Criteria | Restructuring / Hardship completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## REF-001 — Refund

| Field                | Specification                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | REF-001                                                                                                                                                                                                                     |
| Feature Name         | Refund                                                                                                                                                                                                                      |
| Purpose              | Reverse eligible purchase value and adjust obligations.                                                                                                                                                                     |
| Actors               | Customer / Merchant / Finance                                                                                                                                                                                               |
| Priority             | MVP                                                                                                                                                                                                                         |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                    |
| Main Flow            | User initiates refund → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                           |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                               |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.       |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                    |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                           |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                    |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                     |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                   |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                              |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                            |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                     |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                             |
| Acceptance Criteria  | Refund completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## REF-002 — Partial Refund

| Field         | Specification                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID    | REF-002                                                                                                                                                   |
| Feature Name  | Partial Refund                                                                                                                                            |
| Purpose       | Adjust a portion without corrupting schedule or ledger.                                                                                                   |
| Actors        | Customer / Merchant / Finance                                                                                                                             |
| Priority      | MVP                                                                                                                                                       |
| Preconditions | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled |



|                      |                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | when configured and confirmed.                                                                                                                                                                                                      |
| Main Flow            | User initiates partial refund → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                   |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                       |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.               |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                            |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                   |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                            |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                             |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                           |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                      |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                    |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                             |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                     |
| Acceptance Criteria  | Partial Refund completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## SET-001 — Merchant Settlement

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | SET-001                                                                                                                                                                                                               |
| Feature Name         | Merchant Settlement                                                                                                                                                                                                   |
| Purpose              | Pay merchant according to agreed model and track status.                                                                                                                                                              |
| Actors               | Merchant / Finance                                                                                                                                                                                                    |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates merchant settlement → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                              |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |

|                     |                                                                                                                                                                                                                                          |
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| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                           |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                         |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                  |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                          |
| Acceptance Criteria | Merchant Settlement completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## FIN-001 — Fees

| Field                | Specification                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | FIN-001                                                                                                                                                                                                                   |
| Feature Name         | Fees                                                                                                                                                                                                                      |
| Purpose              | Configure lawful fees without hard-coding.                                                                                                                                                                                |
| Actors               | Finance / Admin                                                                                                                                                                                                           |
| Priority             | MVP                                                                                                                                                                                                                       |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                  |
| Main Flow            | User initiates fees → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                           |
| Acceptance Criteria  | Fees completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## HIS-001 — Customer History

| Field        | Specification                                     |
|--------------|---------------------------------------------------|
| Feature ID   | HIS-001                                           |
| Feature Name | Customer History                                  |
| Purpose      | Show and use verified purchase/repayment history. |

|                      |                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors               | Customer / Risk                                                                                                                                                                                                                       |
| Priority             | MVP                                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                              |
| Main Flow            | User initiates customer history → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                 |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria  | Customer History completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## RPT-001 — Merchant Reports

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | RPT-001                                                                                                                                                                                                               |
| Feature Name         | Merchant Reports                                                                                                                                                                                                      |
| Purpose              | Provide sales, BNPL, refund, fee and settlement reporting.                                                                                                                                                            |
| Actors               | Merchant                                                                                                                                                                                                              |
| Priority             | Should                                                                                                                                                                                                                |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates merchant reports → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |

|                     |                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Interface      | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications       | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria | Merchant Reports completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## RPT-002 — Customer Reports

| Field                | Specification                                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | RPT-002                                                                                                                                                                                                                               |
| Feature Name         | Customer Reports                                                                                                                                                                                                                      |
| Purpose              | Provide purchases, payments, due dates and statements.                                                                                                                                                                                |
| Actors               | Customer                                                                                                                                                                                                                              |
| Priority             | MVP                                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                              |
| Main Flow            | User initiates customer reports → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                 |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                       |
| Acceptance Criteria  | Customer Reports completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## RSK-001 — Fraud Detection

| Field                | Specification                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | RSK-001                                                                                                                                                                                                                              |
| Feature Name         | Fraud Detection                                                                                                                                                                                                                      |
| Purpose              | Detect identity, account, merchant, transaction and refund fraud.                                                                                                                                                                    |
| Actors               | Risk / Compliance                                                                                                                                                                                                                    |
| Priority             | MVP                                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                             |
| Main Flow            | User initiates fraud detection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                    |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                        |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                             |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                    |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                             |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                              |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                            |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                       |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                     |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                              |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                      |
| Acceptance Criteria  | Fraud Detection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## DSP-001 — Disputes

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | DSP-001                                                                                                                                                                                  |
| Feature Name      | Disputes                                                                                                                                                                                 |
| Purpose           | Investigate financial and merchant disputes with evidence.                                                                                                                               |
| Actors            | Customer / Merchant / Support                                                                                                                                                            |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates disputes → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.            |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                        |
| Failure Flows     | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                            |
| Business Rules    | Rules are configuration/version controlled; no financial state changes without                                                                                                           |

|                      |                                                                                                                                                                                                                               |
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|                      | traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                                                                                        |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                      |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                             |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                      |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                       |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                     |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                              |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                       |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                               |
| Acceptance Criteria  | Disputes completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## SUP-001 — Customer Support

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | SUP-001                                                                                                                                                                                                               |
| Feature Name         | Customer Support                                                                                                                                                                                                      |
| Purpose              | Track assistance, attachments, escalation and resolution.                                                                                                                                                             |
| Actors               | Customer / Support                                                                                                                                                                                                    |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates customer support → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                      |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                               |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                       |

|                     |                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptance Criteria | Customer Support completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ADM-001 — Admin Platform

| Field                | Specification                                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | ADM-001                                                                                                                                                                                                                             |
| Feature Name         | Admin Platform                                                                                                                                                                                                                      |
| Purpose              | Operate customers, merchants, finance, risk, support and configuration.                                                                                                                                                             |
| Actors               | Admin Roles                                                                                                                                                                                                                         |
| Priority             | MVP                                                                                                                                                                                                                                 |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                            |
| Main Flow            | User initiates admin platform → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                   |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                       |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.               |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                            |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                   |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                            |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                             |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                           |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                      |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                    |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                             |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                     |
| Acceptance Criteria  | Admin Platform completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## IAM-001 — Roles & Permissions

| Field             | Specification                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID        | IAM-001                                                                                                                                                                                  |
| Feature Name      | Roles & Permissions                                                                                                                                                                      |
| Purpose           | Apply least privilege and separation of duties.                                                                                                                                          |
| Actors            | All privileged roles                                                                                                                                                                     |
| Priority          | MVP                                                                                                                                                                                      |
| Preconditions     | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |
| Main Flow         | User initiates roles & permissions → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications. |
| Alternative Flows | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only                                       |

|                      |                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | when approved.                                                                                                                                                                                                                           |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                            |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                    |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                 |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                        |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                 |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                  |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                           |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                         |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                  |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                          |
| Acceptance Criteria  | Roles & Permissions completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## AUD-001 — Audit Logs

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | AUD-001                                                                                                                                                                                                               |
| Feature Name         | Audit Logs                                                                                                                                                                                                            |
| Purpose              | Record significant financial/admin state changes immutably.                                                                                                                                                           |
| Actors               | Compliance / Audit                                                                                                                                                                                                    |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates audit logs → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                       |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                        |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID,                                                                                                                                           |



|                     |                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | policy/config version and timestamp.                                                                                                                                                                                            |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                         |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                 |
| Acceptance Criteria | Audit Logs completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## ENT-001 — Enterprise Merchant Connection

| Field                | Specification                                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | ENT-001                                                                                                                                                                                                                                             |
| Feature Name         | Enterprise Merchant Connection                                                                                                                                                                                                                      |
| Purpose              | Connect approved enterprise POS/e-commerce systems.                                                                                                                                                                                                 |
| Actors               | Enterprise Merchant                                                                                                                                                                                                                                 |
| Priority             | Should                                                                                                                                                                                                                                              |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                            |
| Main Flow            | User initiates enterprise merchant connection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                                   |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                                       |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                               |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                            |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                                   |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                            |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                             |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                           |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                                      |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                                    |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                             |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                                     |
| Acceptance Criteria  | Enterprise Merchant Connection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## RPT-003 — Management Reporting

| Field         | Specification                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID    | RPT-003                                                                                                                                                                                  |
| Feature Name  | Management Reporting                                                                                                                                                                     |
| Purpose       | Track GMV, repayment, overdue, fraud, revenue and operational health.                                                                                                                    |
| Actors        | Management                                                                                                                                                                               |
| Priority      | MVP                                                                                                                                                                                      |
| Preconditions | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed. |

|                      |                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main Flow            | User initiates management reporting → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria  | Management Reporting completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CFG-001 — System Configuration

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CFG-001                                                                                                                                                                                                               |
| Feature Name         | System Configuration                                                                                                                                                                                                  |
| Purpose              | Version and approve plans, limits, fees, provider settings and policies.                                                                                                                                              |
| Actors               | Authorized Admin                                                                                                                                                                                                      |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates system configuration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                             |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                              |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                               |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                             |

|                     |                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria | System Configuration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CMP-001 — Terms & Consent

| Field                | Specification                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | CMP-001                                                                                                                                                                                                                              |
| Feature Name         | Terms & Consent                                                                                                                                                                                                                      |
| Purpose              | Record exact terms/offer/privacy consent versions.                                                                                                                                                                                   |
| Actors               | Customer / Compliance                                                                                                                                                                                                                |
| Priority             | MVP                                                                                                                                                                                                                                  |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                             |
| Main Flow            | User initiates terms & consent → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                    |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                        |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                             |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                    |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                             |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                              |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                            |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                       |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                     |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                              |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                      |
| Acceptance Criteria  | Terms & Consent completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## CMP-002 — Complaint Management

| Field        | Specification                                              |
|--------------|------------------------------------------------------------|
| Feature ID   | CMP-002                                                    |
| Feature Name | Complaint Management                                       |
| Purpose      | Manage complaints, SLA evidence and regulatory escalation. |
| Actors       | Customer / Support / Compliance                            |

|                      |                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority             | MVP                                                                                                                                                                                                                                       |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                  |
| Main Flow            | User initiates complaint management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                         |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                             |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                     |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                  |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                         |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                  |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                   |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                 |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                            |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                          |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                   |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                           |
| Acceptance Criteria  | Complaint Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## REC-001 — Reconciliation

| Field                | Specification                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | REC-001                                                                                                                                                                                                               |
| Feature Name         | Reconciliation                                                                                                                                                                                                        |
| Purpose              | Compare provider, bank, settlement and internal ledger records.                                                                                                                                                       |
| Actors               | Finance / Operations                                                                                                                                                                                                  |
| Priority             | MVP                                                                                                                                                                                                                   |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                              |
| Main Flow            | User initiates reconciliation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                   |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                     |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                         |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success. |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                              |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                     |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending                                                                                                                                         |

|                     |                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | states, accessible error and support path.                                                                                                                                                                                          |
| Notifications       | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                             |
| Security            | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                           |
| Financial Impact    | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                      |
| Audit Requirements  | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                    |
| Data Requirements   | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                             |
| Dependencies        | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                     |
| Acceptance Criteria | Reconciliation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## SEC-001 — Device & Session Security

| Field                | Specification                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature ID           | SEC-001                                                                                                                                                                                                                                        |
| Feature Name         | Device & Session Security                                                                                                                                                                                                                      |
| Purpose              | Protect sessions, devices, tokens and high-risk actions.                                                                                                                                                                                       |
| Actors               | All users                                                                                                                                                                                                                                      |
| Priority             | MVP                                                                                                                                                                                                                                            |
| Preconditions        | Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.                                                       |
| Main Flow            | User initiates device & session security → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.                                                 |
| Alternative Flows    | Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.                                                                              |
| Failure Flows        | Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.                                                                                  |
| Business Rules       | Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.                          |
| Required Information | Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.                                                                                                       |
| Validation           | Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.                                                                                              |
| User Interface       | Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.                                                                                                                       |
| Notifications        | Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.                                                                                                                                        |
| Security             | Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.                                                                                      |
| Financial Impact     | If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.                                                 |
| Audit Requirements   | Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.                                                                                                                               |
| Data Requirements    | Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.                                                                                                        |
| Dependencies         | Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.                                                                                                                                |
| Acceptance Criteria  | Device & Session Security completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state. |

## 4. Traceability

| Feature ID | Related Business Requirements          |
|------------|----------------------------------------|
| AUTH-001   | See BRD/SRS cross-cutting requirements |
| AUTH-002   | See BRD/SRS cross-cutting requirements |
| AUTH-003   | See BRD/SRS cross-cutting requirements |
| CUS-001    | See BRD/SRS cross-cutting requirements |
| KYC-001    | See BRD/SRS cross-cutting requirements |
| KYC-002    | See BRD/SRS cross-cutting requirements |
| MER-001    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| MER-002    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| MER-003    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| MER-004    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| MER-005    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| MER-006    | BR-007, BR-008, BR-009, BR-021, BR-023 |
| QR-001     | See BRD/SRS cross-cutting requirements |
| QR-002     | See BRD/SRS cross-cutting requirements |
| DISC-001   | See BRD/SRS cross-cutting requirements |
| CRD-001    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| CRD-002    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| CRD-003    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| CRD-004    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| CRD-005    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| CRD-006    | BR-002, BR-003, BR-030, BR-034, BR-037 |
| BNPL-001   | See BRD/SRS cross-cutting requirements |
| BNPL-002   | See BRD/SRS cross-cutting requirements |
| BNPL-003   | See BRD/SRS cross-cutting requirements |
| BNPL-004   | See BRD/SRS cross-cutting requirements |
| PAY-001    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-002    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-003    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-004    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-005    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-006    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-007    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-008    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-009    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-010    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| PAY-011    | BR-005, BR-007, BR-012, BR-014, BR-015 |
| COL-001    | See BRD/SRS cross-cutting requirements |
| COL-002    | See BRD/SRS cross-cutting requirements |
| REF-001    | See BRD/SRS cross-cutting requirements |
| REF-002    | See BRD/SRS cross-cutting requirements |
| SET-001    | See BRD/SRS cross-cutting requirements |
| FIN-001    | See BRD/SRS cross-cutting requirements |
| HIS-001    | See BRD/SRS cross-cutting requirements |
| RPT-001    | See BRD/SRS cross-cutting requirements |
| RPT-002    | See BRD/SRS cross-cutting requirements |
| RSK-001    | See BRD/SRS cross-cutting requirements |
| DSP-001    | See BRD/SRS cross-cutting requirements |
| SUP-001    | See BRD/SRS cross-cutting requirements |
| ADM-001    | See BRD/SRS cross-cutting requirements |
| IAM-001    | See BRD/SRS cross-cutting requirements |
| AUD-001    | See BRD/SRS cross-cutting requirements |
| ENT-001    | See BRD/SRS cross-cutting requirements |
| RPT-003    | See BRD/SRS cross-cutting requirements |
| CFG-001    | See BRD/SRS cross-cutting requirements |
| CMP-001    | See BRD/SRS cross-cutting requirements |
| CMP-002    | See BRD/SRS cross-cutting requirements |
| REC-001    | See BRD/SRS cross-cutting requirements |
| SEC-001    | See BRD/SRS cross-cutting requirements |

## References

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Supports: Current licensed/commercialized payment instrument issuers and payment system operators, including Telebirr, Kacha, M-PESA Ethiopia, EthSwitch, ArifPay, Chapa and SantimPay.
- **[ETH-NBE-02] VERIFIED — National Bank of Ethiopia.** Financial Stability Report — March 2026.  
[https://nbe.gov.et/wp-content/uploads/2026/03/Financial-Stability-Report\\_MARCH\\_2026.pdf](https://nbe.gov.et/wp-content/uploads/2026/03/Financial-Stability-Report_MARCH_2026.pdf)  
Supports: Current payment-sector regulatory framework, digital-finance context and systemic risk context.
- **[ETH-NBE-03] VERIFIED — National Bank of Ethiopia.** National Digital Payments Strategy 2026–2030 — Draft.  
[https://nbe.gov.et/wp-content/uploads/2025/12/Ethiopia\\_NDPS\\_Draft\\_F.pdf](https://nbe.gov.et/wp-content/uploads/2025/12/Ethiopia_NDPS_Draft_F.pdf)  
Supports: Interoperability direction and digital-payment strategy context; document is a draft.
- **[ETH-NBE-04] VERIFIED — National Bank of Ethiopia.** Financial Consumer Protection — Complaint Handling.  
<https://nbe.gov.et/fcpe/complaint/>  
Supports: Complaint escalation after first contacting the financial service provider; NBE review where unresolved/no response within 10 business days.
- **[ETH-DP-01] VERIFIED — Federal Democratic Republic of Ethiopia / Ministry of Justice.** Personal Data Protection Proclamation No. 1321/2024.  
<https://justice.gov.et/en/law/personal-data-protection-proclamation/>  
Supports: Ethiopian personal-data protection framework.
- **[ETH-QR-01] VERIFIED — EthSwitch.** Interoperable QR Standard.  
<https://ethswitch.com/wp-content/uploads/2024/12/Interoperable-QR-Standard.pdf>  
Supports: Interoperable QR design direction for merchant payments and limitations of closed-loop QR.
- **[TEL-01] VERIFIED — Ethio telecom.** Telebirr Developer Portal — Getting Started.  
<https://developer.ethiotelcom.et/docs/GettingStarted>  
Supports: Publicly documented Telebirr business integration categories including C2B, B2B, H5, Mini App and subscription payment.
- **[TEL-02] VERIFIED — Ethio telecom.** Telebirr H5 C2B Web Payment Integration.  
<https://developer.ethiotelcom.et/docs/category/h5-c2b-web-payment-integration>  
Supports: Public create-order, checkout, query/status, callback and refund integration documentation.
- **[TEL-03] VERIFIED — Ethio telecom.** Telebirr Schedule Payment / CreateDisburseOrder.  
<https://developer.ethiotelcom.et/docs/Schedule%20Payment/CreateDisburseOrder>  
Supports: Public documentation for mandate-based scheduled deduction; BNPL repayment use requires commercial/regulatory confirmation.
- **[CBE-01] VERIFIED — Commercial Bank of Ethiopia.** CBE Birr.  
<https://combanketh.et/ways-of-banking/cbe-birr>  
Supports: Official description of CBE Birr customer/mobile-money services.
- **[CBE-02] VERIFIED — Commercial Bank of Ethiopia.** CBE Products and Services.  
[https://combanketh.et/cbeapi/uploads/CBE\\_Products\\_and\\_services\\_English\\_84e511c740.pdf](https://combanketh.et/cbeapi/uploads/CBE_Products_and_services_English_84e511c740.pdf)  
Supports: CBE Birr supports transfers, cash out, airtime, goods purchase and bill payment; no equivalent public direct BNPL integration specification verified.
- **[MPE-01] VERIFIED — Safaricom Ethiopia.** M-PESA Ethiopia Merchant Terms / Business Use.  
<https://www.safaricom.et/>  
Supports: M-PESA Ethiopia supports merchant payments/business arrangements; exact BNPL API onboarding and sandbox require provider confirmation.

- **[QCB-01] VERIFIED — Qatar Central Bank.** Buy Now Pay Later Regulation.  
<https://www.qcb.gov.qa/Services/Financial%20Technology/QCB%20BNPL%20Regulation.pdf>  
Supports: Qatar definition, eligibility and regulated BNPL-provider framework. Qatar law is not Ethiopian law.
- **[QTR-TC] VERIFIED — PayLater Qatar.** PayLater General Terms and Conditions — client-provided 20-page copy.  
Client-provided PDF reviewed 21 August 2026  
Supports: Customer/merchant/PayLater roles, eligibility, 25% first payment, PayMore, limits, repayment, default, restructuring, disputes, data handling and account rules.
- **[QTR-PL-01] VERIFIED — PayLater Qatar.** PayLater Terms and Conditions.  
<https://paylaterapp.com/terms-and-conditions/>  
Supports: Current public PayLater Qatar contractual product model.
- **[QTR-PL-02] VERIFIED — PayLater Qatar.** How can I sign up for PayLater?.  
<https://help.paylaterapp.com/en/support/solutions/articles/154000139292-how-can-i-sign-up-for-paylater->  
Supports: Phone OTP, QID capture and liveness-based customer onboarding.
- **[QTR-PL-03] VERIFIED — PayLater Qatar.** What is my credit limit and how much can I spend?.  
<https://help.paylaterapp.com/en/support/solutions/articles/154000139303-what-is-my-credit-limit-and-how-much-can-i-spend-with-paylater->  
Supports: Credit limit uses creditworthiness, PayLater repayment history and account usage.
- **[QTR-PL-04] VERIFIED — PayLater Qatar.** Where can I use PayLater?.  
<https://help.paylaterapp.com/en/support/solutions/articles/154000139291-where-can-i-use-paylater->  
Supports: Use across online and in-store merchants.
- **[QTR-CBQ-01] VERIFIED — Commercial Bank Qatar / Vodafone Qatar.** Vodafone Easy-Buy / CBQ Buy Now Pay Later.  
<https://www.vodafone.qa/en/buynowpaylater>  
Supports: Bank-card installment model for online and selected physical stores; full purchase reduces card limit and customer repays equal monthly installments.
- **[QTR-VF-01] VERIFIED — Vodafone Qatar.** Vodafone Partners with Commercial Bank to Offer BNPL.  
<https://www.vodafone.qa/en/about-us/media/press-release/vodafone-partners-with-commercial-bank-to-offer-buy-now-pay-later-installment-plan>  
Supports: Qatar device-installment model through bank partnership and merchant stores.